



Evaluation of Bonding Strength in CFRP Sandwich without Adhesive Film

2025 복합재료학회 추계학술대회

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1. Introduction

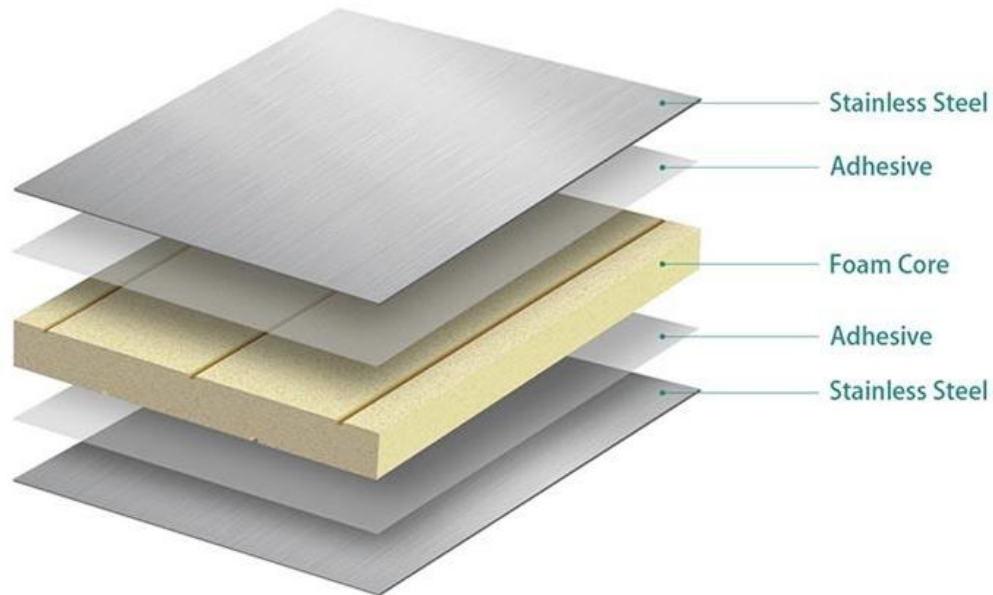
2. Experimental setup

3. Test and results

- Tensile test - ASTM C297/C297M
- Three-point bending test - ASTM C393/C393M
- Ultrasonic evaluation - ASTM E2580
- FEA simulations

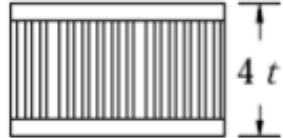
4. Conclusions

1.1 Sandwich composite



< Typical Sandwich structure >



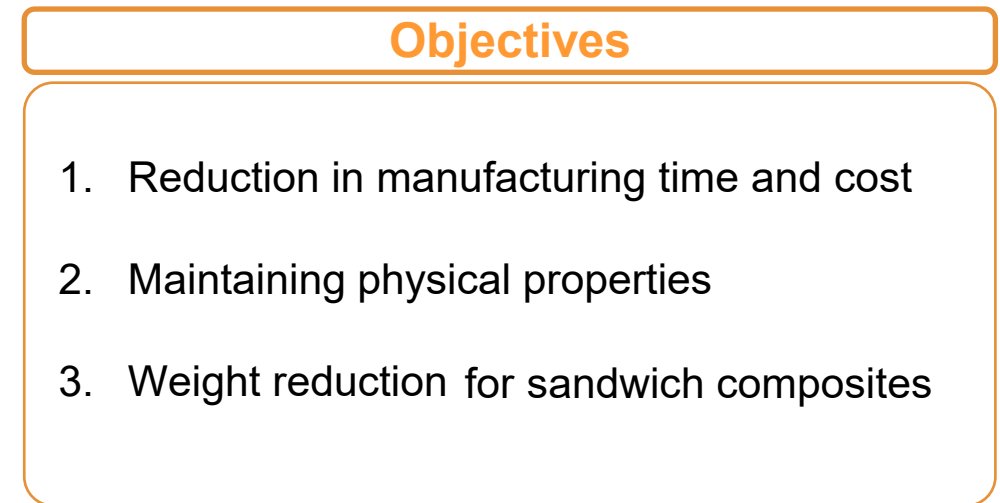
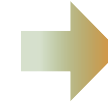
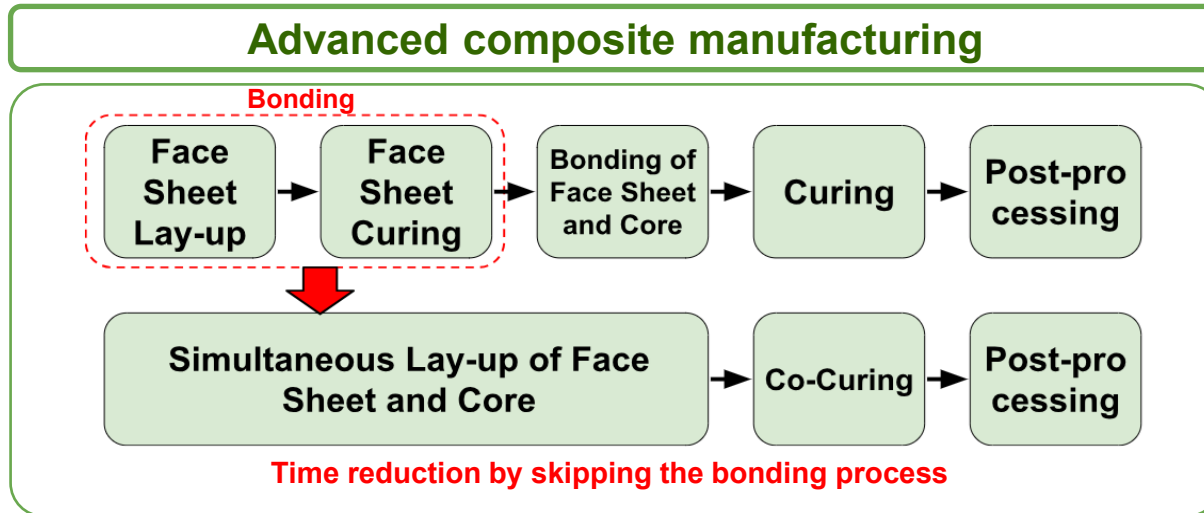
	Solid material	Core thickness t	Core thickness $3t$
			
Bending stiffness	1.0	7.0	37
Bending strength	1.0	3.5	9.2
Weight	1.0	1.03	1.06

< Stiffness and weight analysis of sandwich panels >

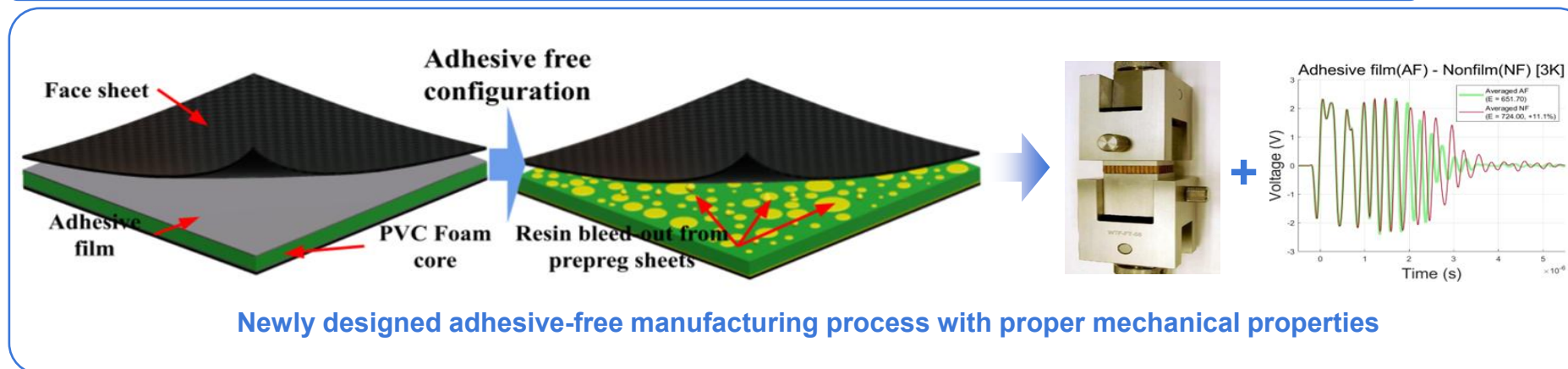
► Sandwich structure holds advantages in lightweightness, stiffness and strength.

1.2 Graphical abstract

► Adhesive-free sandwich composite manufacturing process for time-, cost- and weight-reduction



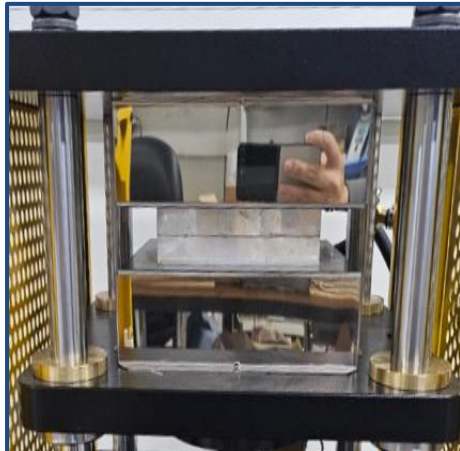
Adhesive film removal / Mechanical testing / NDE



Mechanical testings in this study:
thicknesswise tensile testing and
three-point bending testing

NDE: Non-Destructive Evaluation

2.1 Material properties



	Carbon prepreg			
Property	UD	3K	6 K	12K
Ply pattern	[0] ₆ /core/[0] ₆	[0] ₄ /core/[0] ₄	[0] ₂ /core/[0] ₂	[0] ₂ /core/[0] ₂
Ply thickness	0.15 mm	0.25 mm	0.45 mm	0.45 mm
Pattern	UD	2 × 2 Twill	2 × 2 Twill	2 × 2 Twill
Face sheet thickness	0.9 mm	1 mm	0.9 mm	0.9 mm
Resin weight	37%	42%	38%	38%

Property	AIREX C70.75
Density	80 g/m3
Pattern	Plain
Compressive strength	1.45 N/mm2
Compressive modulus	104 N/mm2
Tensile strength	2.0 N/mm2
Tensile modulus	66 N/mm2
Shear strength	1.2 N/mm2
Shear modulus	30 N/mm2
Ply thickness	5 mm
Resin Weight	-

Property	Adhesive film: GPL-150
Resin weight	150 gsm
Peel Strength (ASTM D3165)	437 N/mm ²
Peel Strength (ASTM D1781)	538 N/mm ²
Tg Storage Modulus Onset	141 °C
Tan Peak	150 °C



<Adhesive Film(AF), No Film(NF)>

► Parameters: Type of skin fabric, manufacturing facility, presence of adhesive film

2.2 Sample preparation



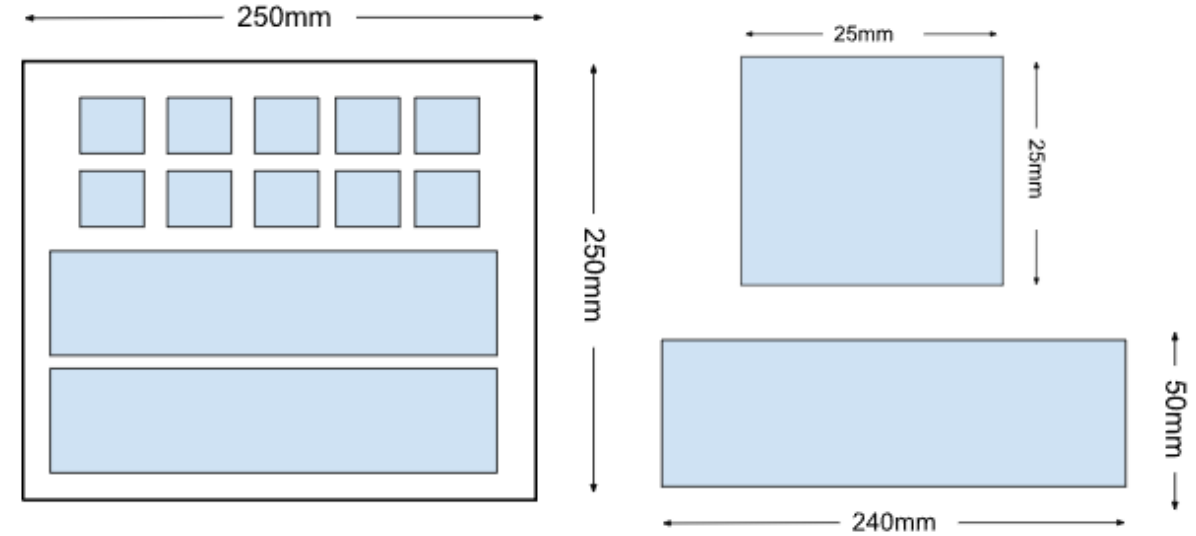
Machining parameters

Tool: Ø2 Carbide Diamond Flat End Mill

Revolution : 8000 RPM

Speed: 400 mm/min

Depth: 0.25-mm-deep * 30 times

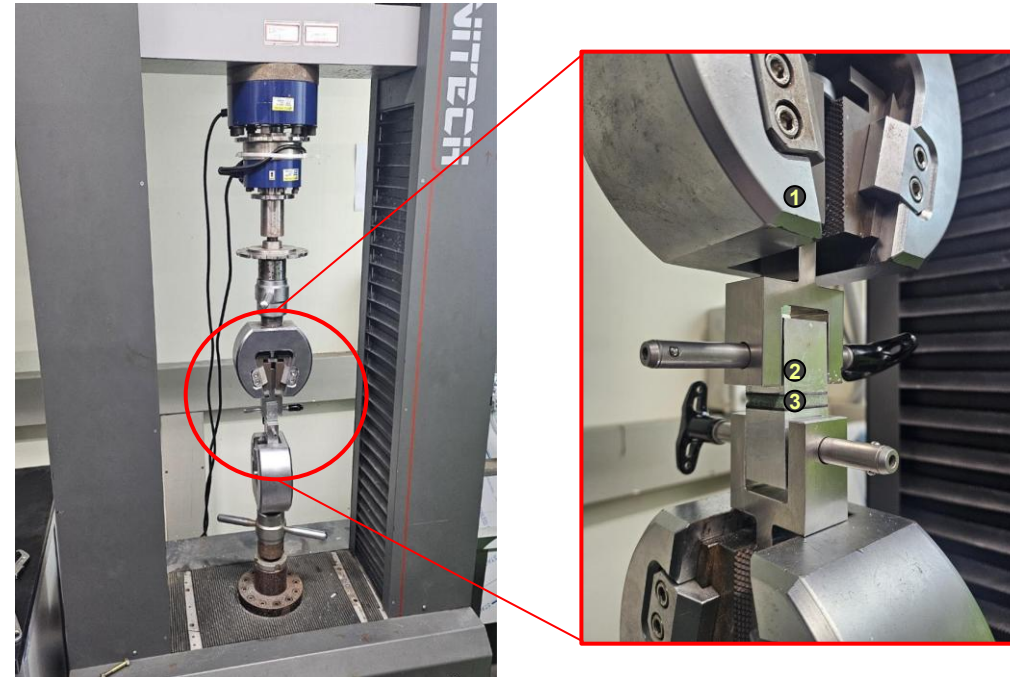
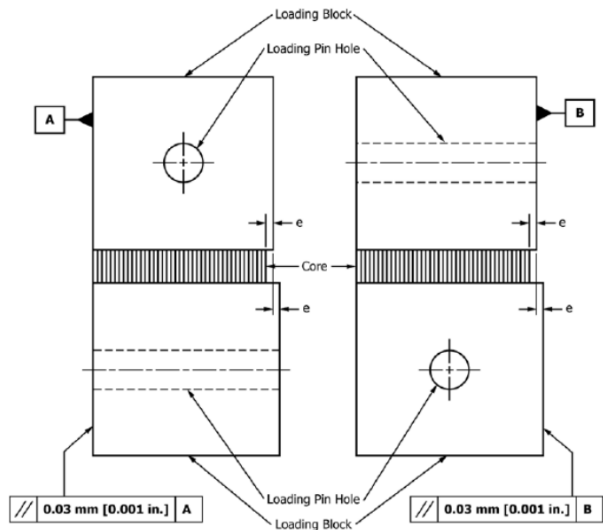
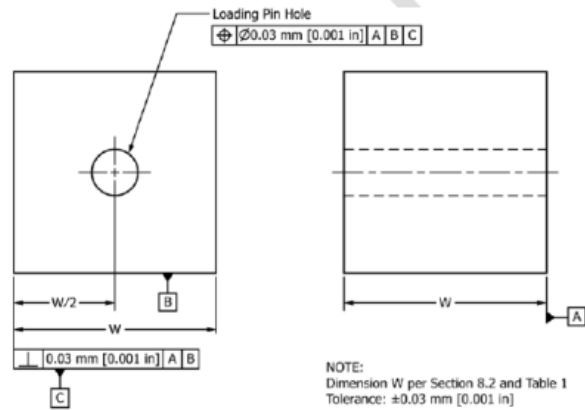


Oven								Hot Press							
Adhesive film (AF)				No - adhesive film (NF)				Adhesive film (AF)				No - adhesive film (NF)			
UD	3k	6k	12k	UD	3k	6k	12k	UD	3k	6k	12k	UD	3k	6k	12k
5EA	5EA	5EA	5EA	5EA	5EA	5EA	5EA	5EA	5EA	5EA	5EA	5EA	5EA	5EA	5EA
Total 80EA															

► Computer numerical control machining for the dimensional precision and consistency

3.1 Testing – Thickness-wise tensile testing

ASTM C297/C297M



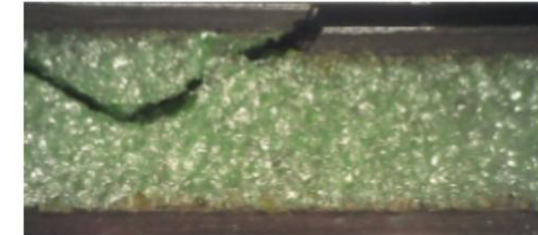
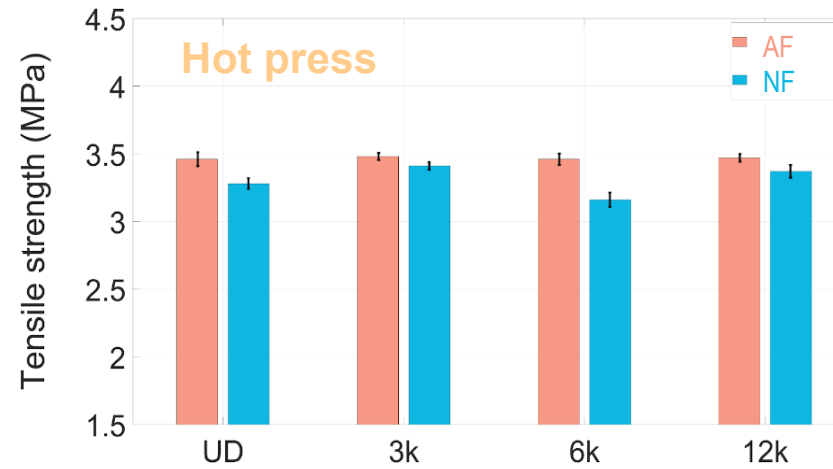
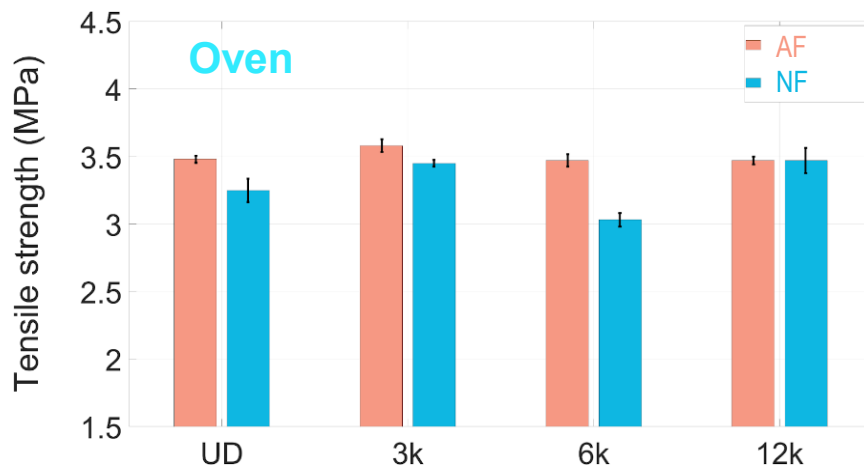
1. Flat jaw jig
2. Loading block
3. Composite sample

Loading rate: 0.5 mm / min

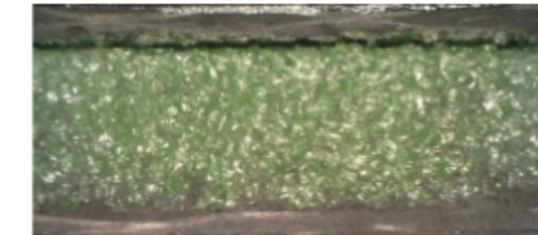
Tensile strength: $\sigma_t = P_{\max} / A$ (MPa)

3.1 Tensile test results - ASTM C297/C297M

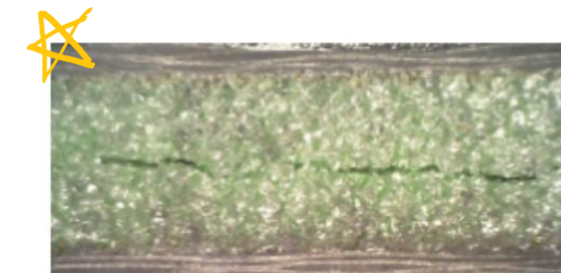
Oven			Hot - press		
Specimen	Film	Failure mode (EA)	Specimen	Film	Failure mode (EA)
UD	X	Skin failure and core failure (2) Core failure and adhesive failure (3)	UD	X	Skin failure and core failure (1) Core failure and adhesive failure (4)
3K	X	Core failure (5)	3K	X	Core failure (5)
6K	X	Core failure and adhesive failure (5)	6K	X	Core failure and adhesive failure (5)
12K	X	Core failure (5)	12K	X	Core failure (5)
UD	O	Core failure (5)	UD	O	Core failure (5)
3K	O	Core failure (5)	3K	O	Core failure (5)
6K	O	Core failure (5)	6K	O	Core failure (5)
12K	O	Core failure (5)	12K	O	Core failure (5)



Skin and core failure



Core and adhesive failure

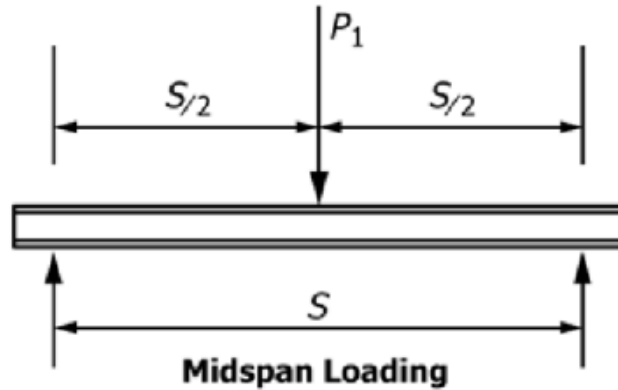


Core failure

► 3K and 12K □ superior with core failure / UD and 6K □ Interfacial failure by insufficient bonding

3.2 Testing – Three-point bending testing

ASTM C393/C393M



(a) 3-Point Loading (Standard Configuration)

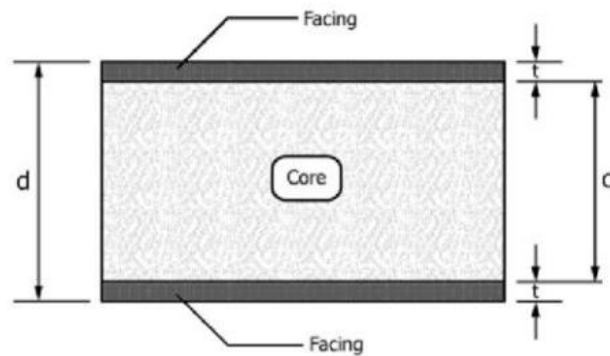
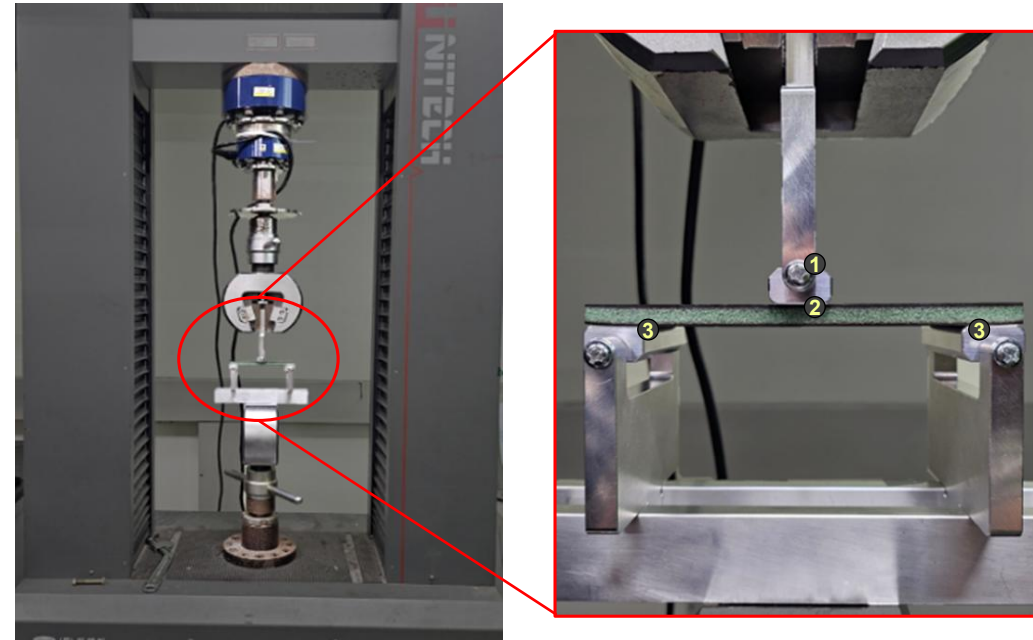


FIG. 2 Sandwich Panel Thickness Dimensions

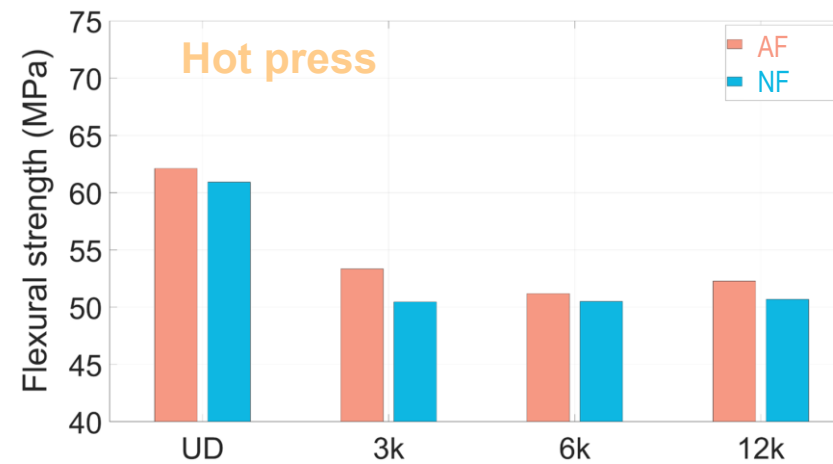
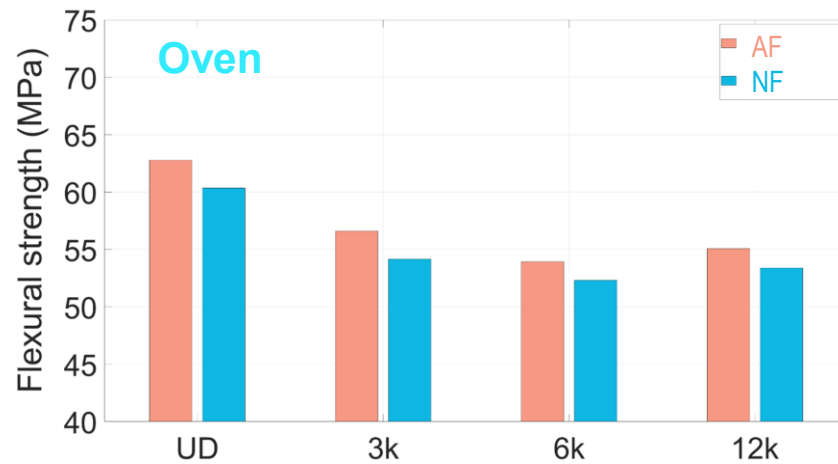
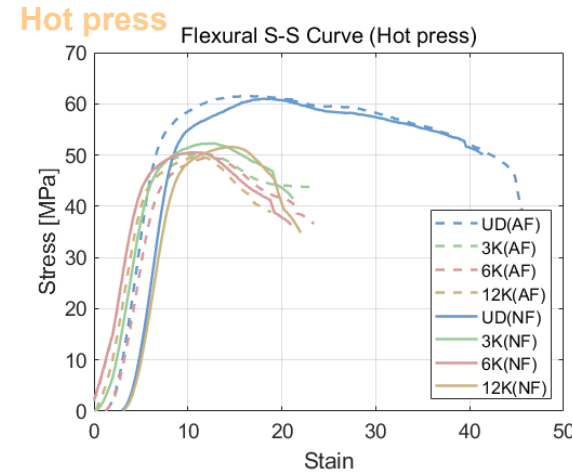
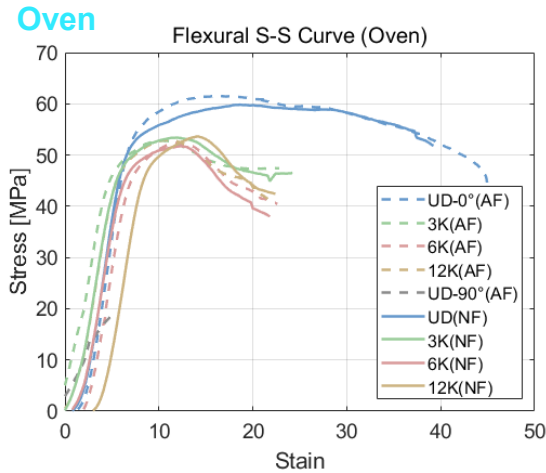


- ❶ Pressing nose
- ❷ Composite sample
- ❸ Supports

Loading rate: 6 mm / min

$$\text{Flexural strength: } \sigma_f = \left(\frac{FL}{4}\right)\left(\frac{d}{2}\right)\left(\frac{12}{bd^3}\right) = \frac{3FL}{2bd^2}$$

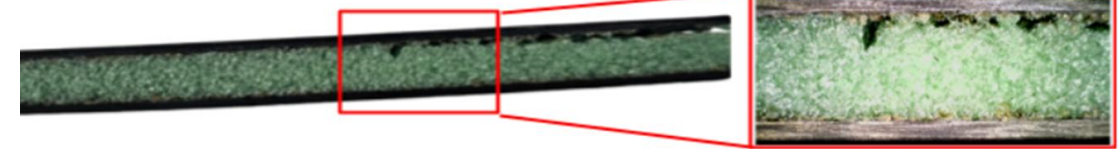
3.2 Three point bending results - ASTM C393/C393M



Top skin failure of 12K (Twill)



Interfacial delamination of UDNF 0° ↔



Combined failure of UDNF 90° ⊙



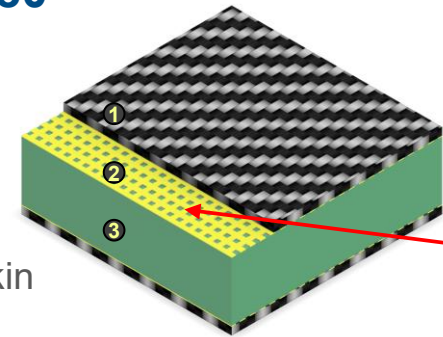
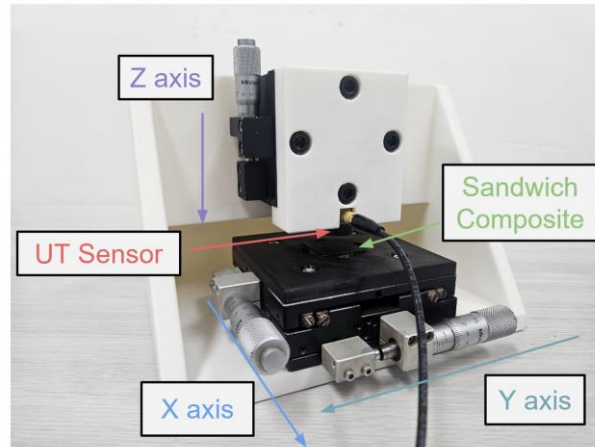
► Not only normal but also shear properties were sufficiently good.

► Thickness-wise compression enabled co-curing by either vacuum or hot-pressing.

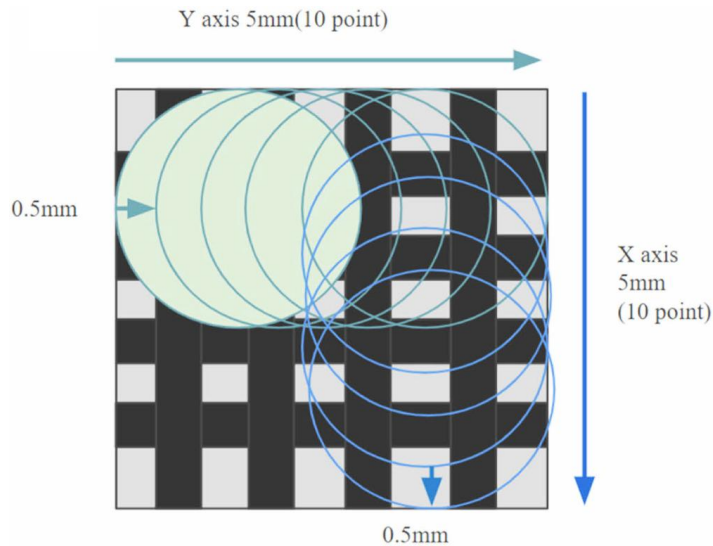
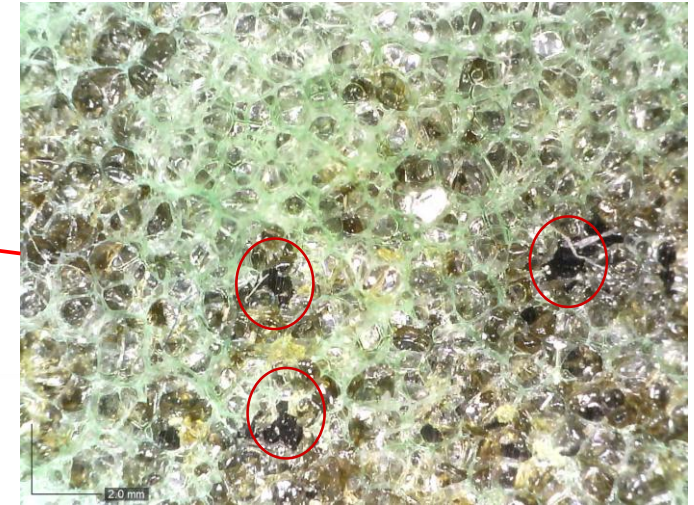
AF: Adhesive Film
NF: Non-adhesive Film
UD: Uni-Directional carbon fiber

► Skin failure was dominant: No problem at the skin-core interface → Well adhered !

3.3 Ultrasonic non-destructive evaluation - ASTM E2580



- ① CFRP skin
- ② Interface
- ③ Foam core



$$X_{avg}[k] = \sum_{n=0}^{M-1} x_{avg}[n] \cdot e^{-j\frac{2\pi}{M}kn} \dots\dots\dots \text{FFT}$$

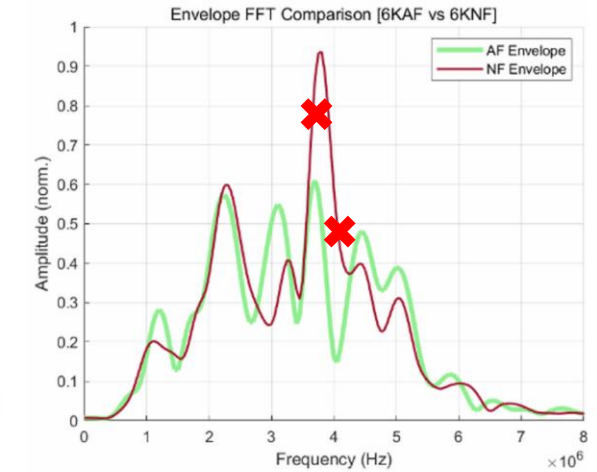
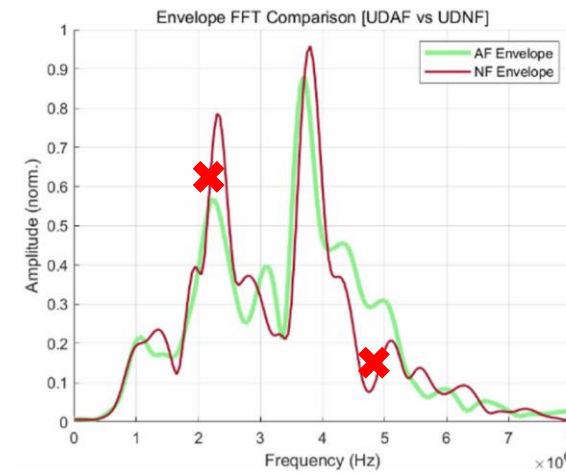
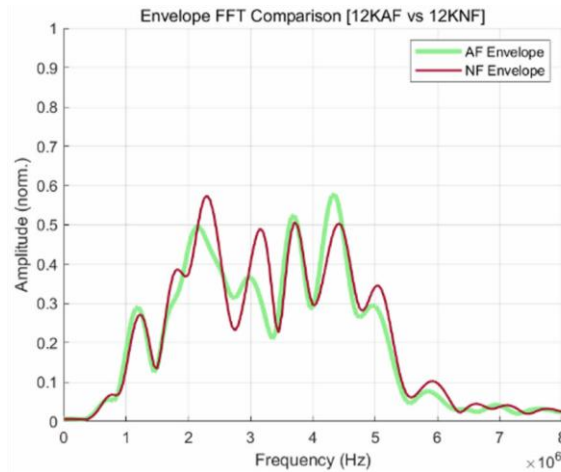
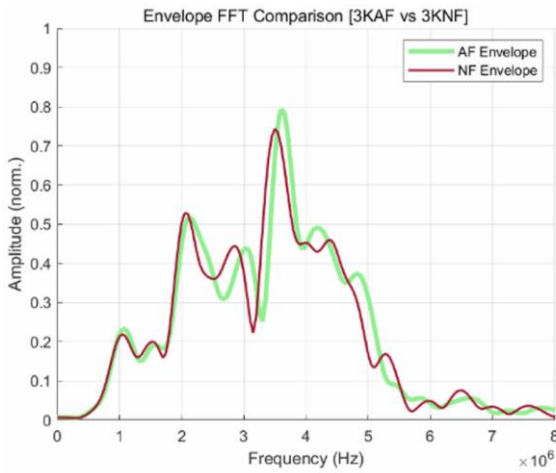
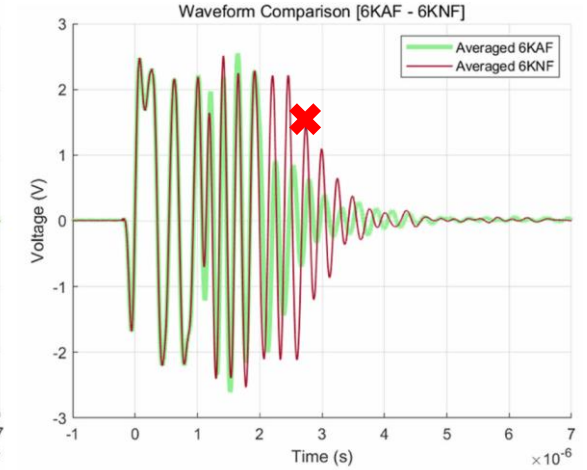
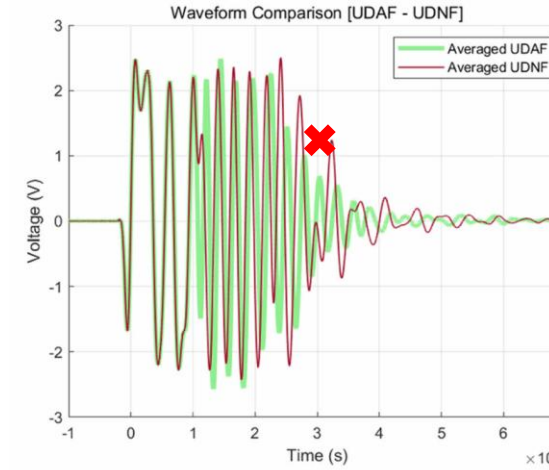
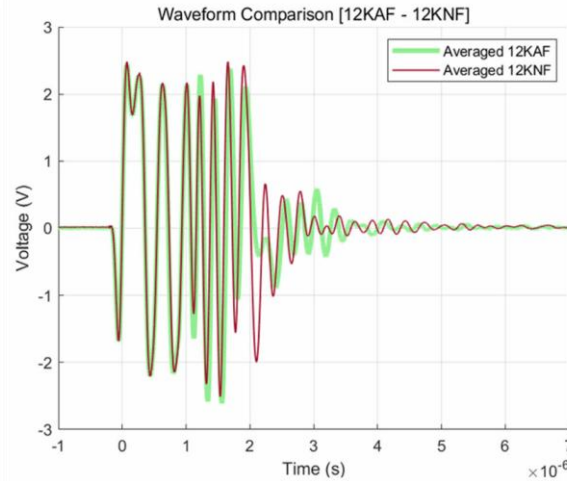
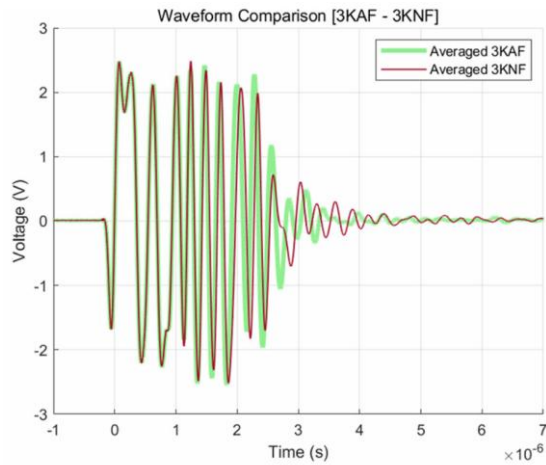
$$x_{avg}(t) = \frac{1}{N} \sum_{i=1}^N x_i(t) \dots\dots\dots \text{Waveform}$$

$$E_t = \int_0^T x_{avg}^2(t) dt \dots\dots\dots \text{Energy}$$

NDE: Non-Destructive Evaluation
FFT: Fast Fourier Transform

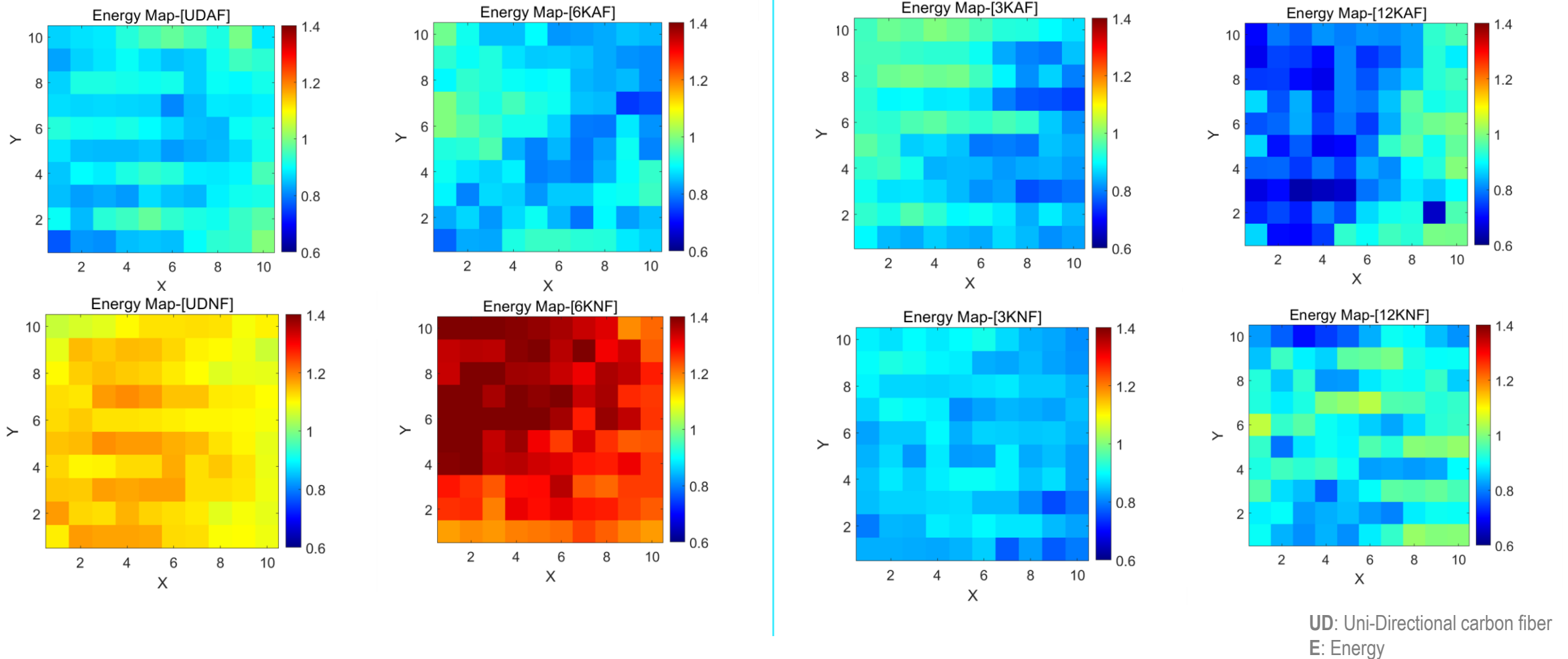
► Ultrasonic NDE was performed with the principles of FFT and energy-based defect detection.

3.3 Ultrasonic non-destructive evaluation - ASTM E2580



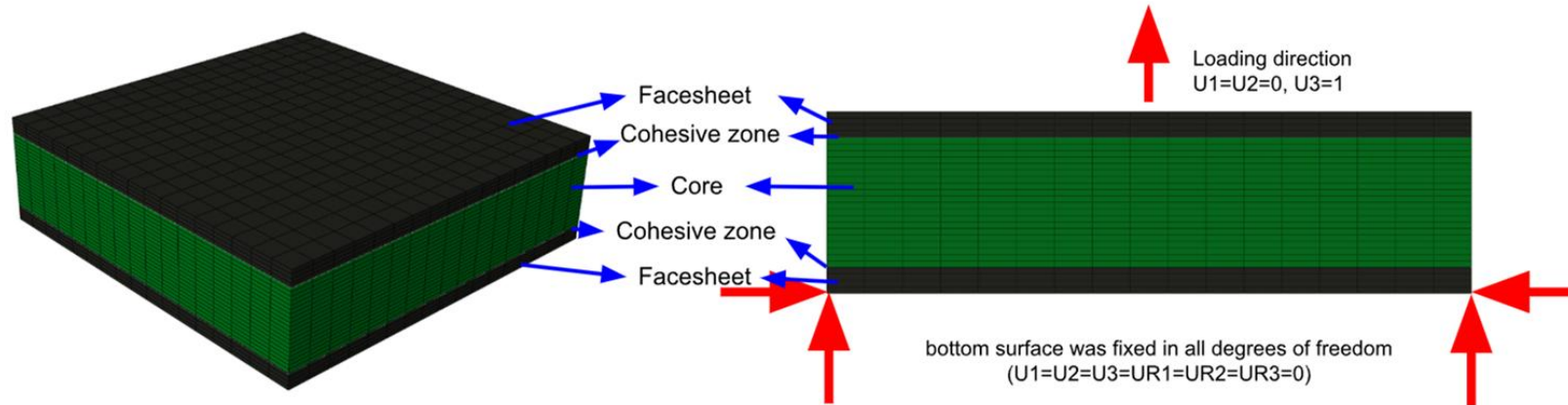
► Nice matching for 3K and 12K / Mismatching for UD and 6K due to bonding defects

3.3 Ultrasonic non-destructive evaluation - ASTM E2580



► Uniform E transfer for 3K and 12K / Non-uniform for UD and 6K due to bonding defects

3.4 FEA simulations



Tool : ABAQUS UMAT

Model size : 25*25*7 (1:1 size)

Mesh size : 1.5*1.5*0.25 mm

Total number of elements : 8092

Mechanical properties of CFRP face sheet

E_1	55.2 GPa
$E_2 = E_3$	7.0 GPa
ν_{12}	0.3
G_{12}	21.0 GPa
ρ	1600 kg/ m ³
X_t	645 MPa
X_c	615 MPa
Y_t	60 MPa
Y_c	200 MPa
S_{12}	69.8 MPa

Mechanical properties of the cohesive layer

K_n	1000 MPa
$K_s = K_t$	1000 MPa
G_n^c	0.35 N/mm
$G_s^c = G_t^c$	0.9 N/mm
τ_n	10 MPa
$\tau_s = \tau_t$	20 MPa
η	1

Mechanical properties of PVC foam core

ρ	80 kg/ m ³
E_c	104 MPa
σ_c	1.45 MPa
G_c	26 MPa
τ_c	1.2 MPa

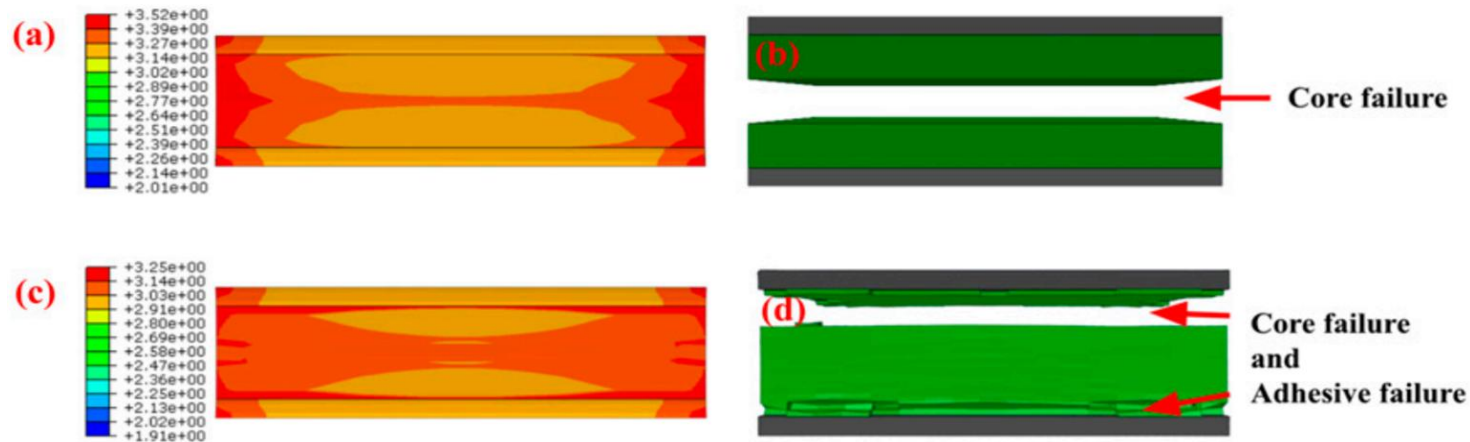
AF: Adhesive Film

NF: Non-adhesive Film

UD: Uni-Directional carbon fiber

► Sandwich composite with cohesive zone modeling was described with finite element model

3.4 FEA simulations



Numerical simulation results of UD specimens under ASTM C297 tensile loading:

- (a) Stress distribution of UD-AF specimen,
- (b) Deformed shape of UD-AF specimen,
- (c) Stress distribution of UD-NF specimen,
- (d) Deformed shape of UD-NF specimen.

Table. Experimental and simulated tensile strength of UD specimens

Specimen	Experimental mean	Simulation	Error rate
UD AF	3.48 MPa	3.52 MPa	1.136%
UD NF	3.30 MPa	3.25 MPa	1.538%

AF: Adhesive Film
 NF: Non-adhesive Film
 UD: Uni-Directional carbon fiber

► Finite element analysis was well investigated representing the core and adhesive failure.

- **3K and 12K face sheets** showed **stable bonding performance** even without an adhesive film, and failure occurred mainly **inside the foam core**, confirming reliable interfacial strength.
- Eliminating the adhesive film **did not significantly reduce mechanical performance**, indicating that **resin infiltration alone can form a sufficient bond** under co-curing.
- The adhesive-free process resulted in a **weight reduction of approximately 5–9%**, contributing to improved lightweight structural efficiency.
- **Manufacturing cost was reduced by about 19–30%**, since the adhesive material and additional bonding steps were removed.
- Overall, the **co-curing process** provides a **simplified, cost-efficient, and lightweight manufacturing method** while maintaining structural reliability.

weight reduction by 25mm²

	Face sheet	Weight AF	Weight NF	Ratio
Oven	3K	2.3067 g	2.1267 g	8.46%
	12K	2.2739 g	2.0797 g	9.34%
Hot press	3K	2.1610 g	2.0349 g	5.75%
	12K	2.0932 g	1.9709 g	6.21%

Manufacturing cost by 1M²

	Face sheet	Cost AF	Cost NF	Ratio
Oven	3K	\$417.81	\$337.50	19.22%
	12K	\$283.61	\$203.29	28.32%
Hot press	3K	\$404.21	\$323.90	19.87%
	12K	\$270.00	\$189.69	29.74%

Thank you for listening



*R*Lab

스마트복합재료연구실

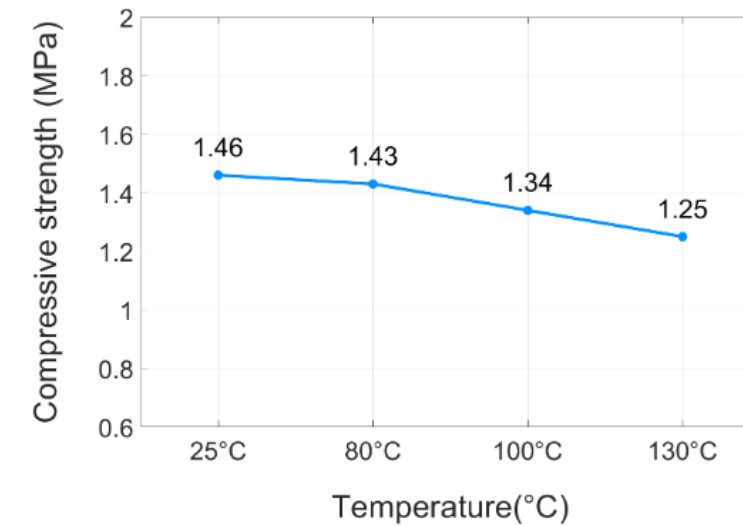
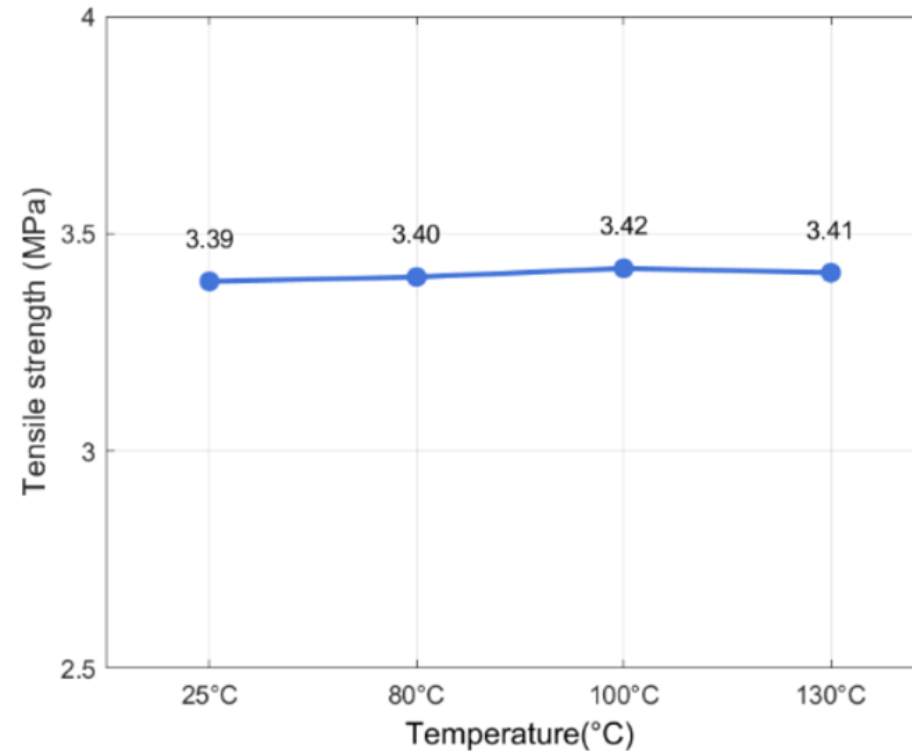
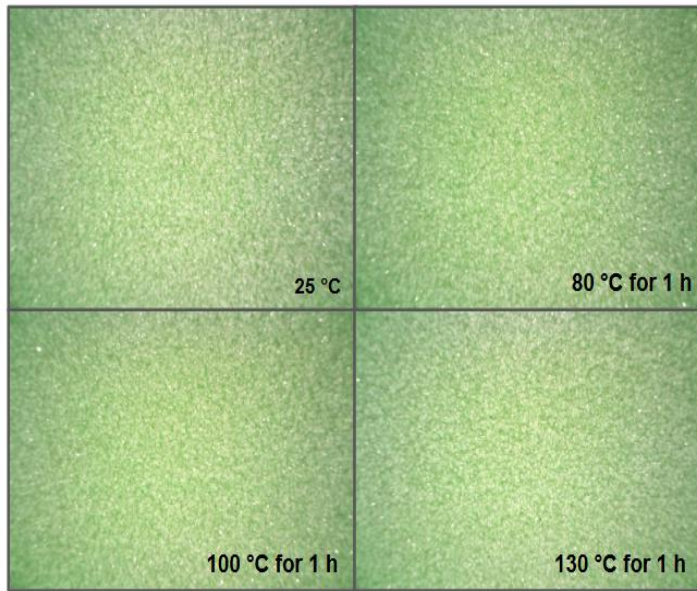


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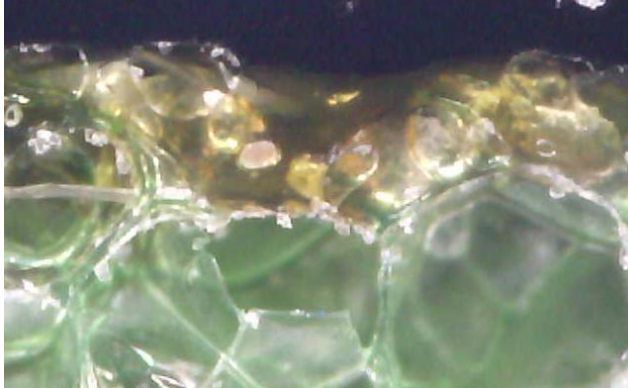
Thermal stability evaluation test:

- (a) PVC foam surface after heat treatment at different temperatures.
- (b) Tensile strength of heat-treated PVC foam.
- (c) compressive strength of heat-treated PVC foam

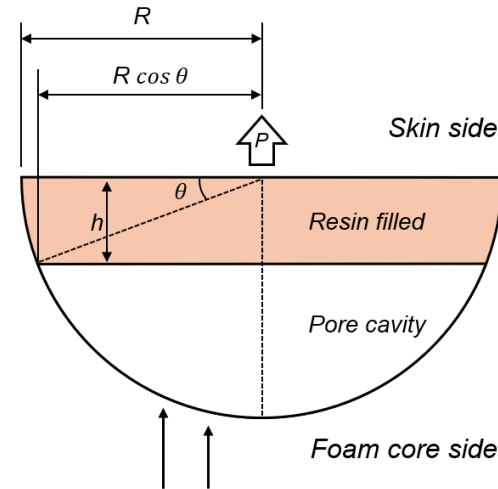


Impregnation analysis of NF specimens

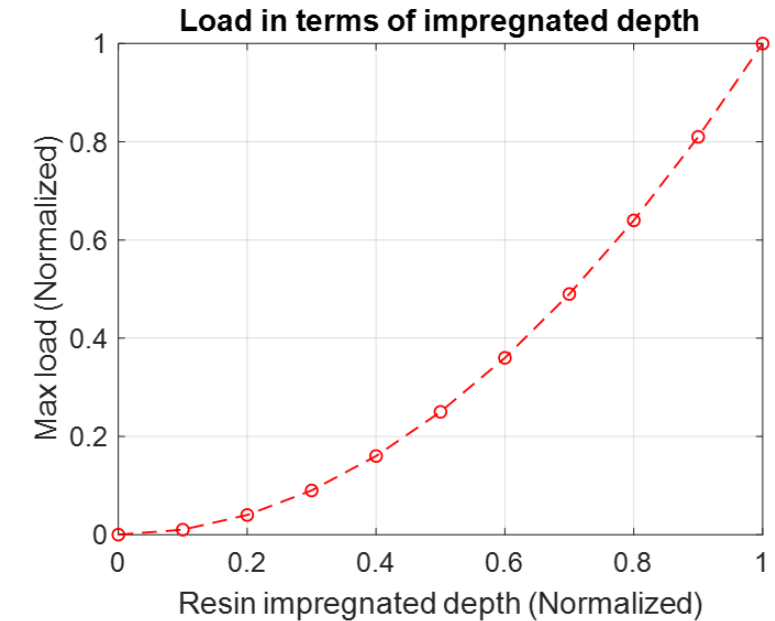
3K NF (well-filled)



(Right) UD NF (poorly filled)



Impregnated depth	$h = R \sin \theta$
Efficient area for bonding	$A_{efficient} = \pi \{R^2 - (R \cos \theta)^2\} = \pi h^2$
Normal joining strength of the adhesive	$\sigma_{joining} = P / A_{efficient}$
Maximum load	$P = \sigma_{joining} * A_{efficient}$
Pore cavity volume	$V_{Pore} = \frac{\pi}{3} \{ (R - h)^2 * (3R - (R - h)) \}$
Volume filled	$V_{Filled} = \frac{2\pi}{3} R^3 - V_{Pore}$
Viscosity and velocity (Newton's law of viscosity)	$\tau_{yx} = \mu * \frac{du}{dy}$



► Resin impregnation is the key for the skin-core adhering force.

NF: Non-adhesive Film
UD: Uni-Directional carbon fiber

► Resin immersed in the prepreg was well impregnated to the foam core: Adhering effect

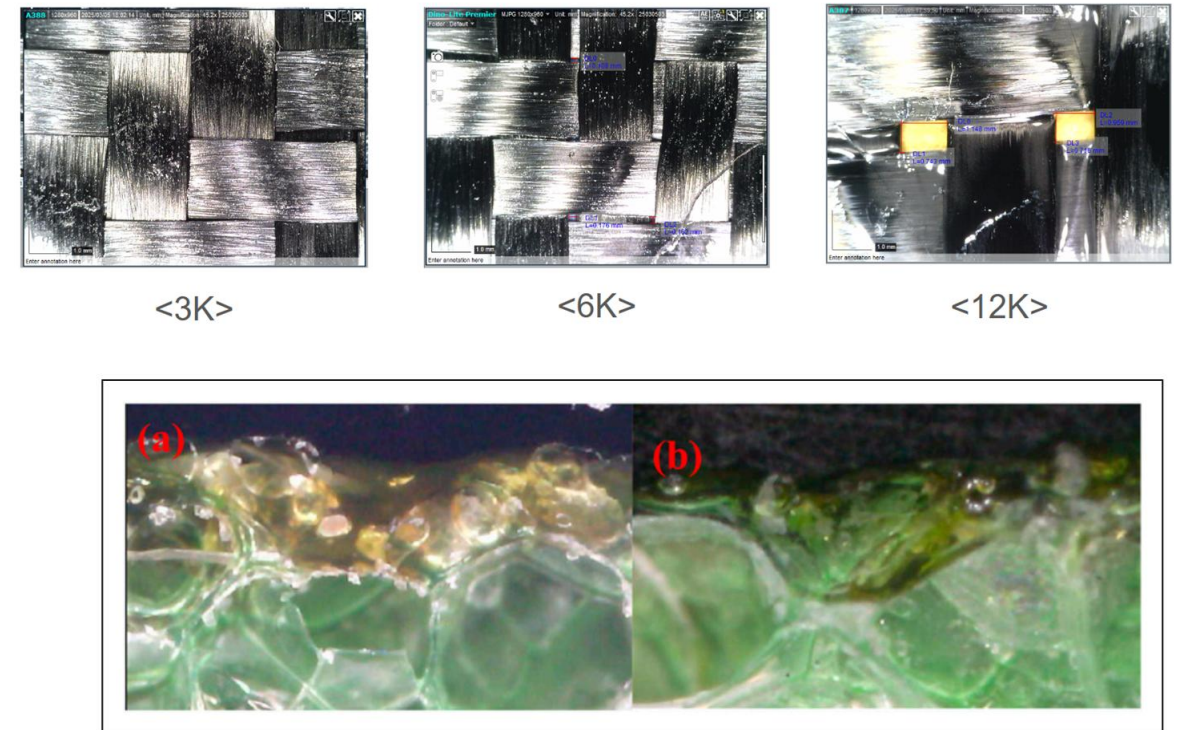
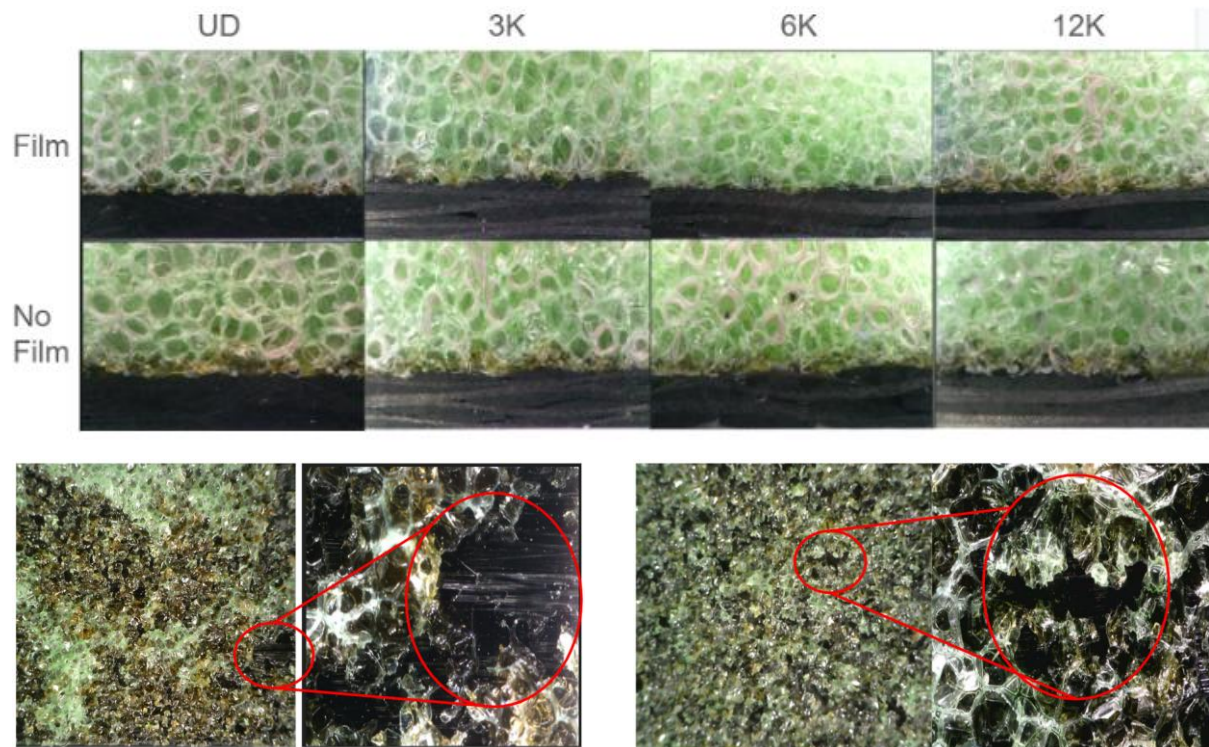
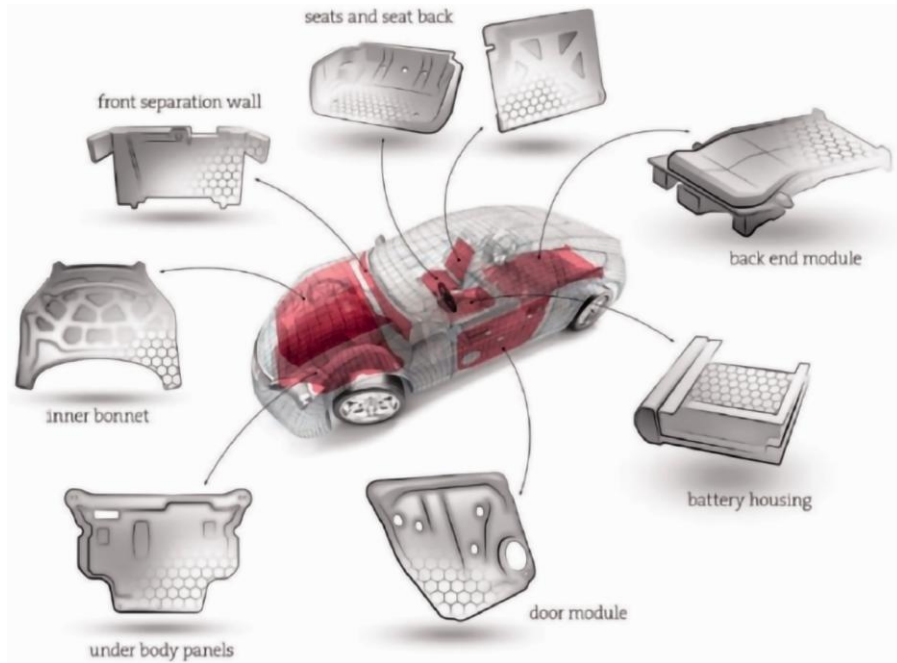


Figure 14. Optical comparison of specimens: (a) 3K NF (well-filled), (b) UD NF (poorly filled).

Larger junction sizes provided more continuous resin flow paths, leading to improved interfacial filling and stronger bonding. In particular, 3K and 12K specimens showed well-filled interfaces with stable adhesion, whereas UD exhibited insufficient resin penetration and local interfacial gaps. The 6K specimen showed partial voids due to non-uniform resin flow, representing an intermediate behavior.

Application



1500kg 의 자동차에서 10%를 감량 할 경우



연비

3.8% ↑



제동거리

5% ↓



가속성능

8% ↑



차체 내구 수명

1.7 배 ↑



배기가스 배출

2.5~8.8% ↓

